

# AYMANE AIT DADS

AI Developer Intern | Generative AI Integration · LangChain · LLM Fine-Tuning & Reasoning

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## PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a 30B-parameter model with LoRA SFT, staged continuation training, and task-level error analysis - directly demonstrating production-grade LLM optimization capability. Fine-tuned a CLIP ViT-L/14 (428M parameters) image-detection model to 0.791 AUC, ranking 1st in the EURECOM class leaderboard and submitting to the NTIRE 2026 @ CVPR CodaBench international benchmark, reducing manual analysis time by 60% at Orange Maroc through automated ML pipelines. Brings hands-on experience with LangChain, OpenAI-compatible APIs, Hugging Face Transformers, RAG, agentic pipelines, and generative AI integration - matching directly the evaluation and optimization scope of this role. Aims to contribute LLM reasoning expertise and ablation-driven engineering discipline to Trusteq's AI component development, helping build robust, production-ready generative AI solutions.

## CORE COMPETENCIES

Generative AI Integration | LangChain Framework | API Integration (OpenAI / Ollama) | LLM Evaluation & Optimization | Data Analysis & Preparation | Agentic Pipelines | RAG & Prompt Engineering | Software with AI Components

## WORK EXPERIENCE

### Orange Maroc

Jul 2024 - Aug 2024

**Data Science Intern** - Casablanca, Morocco

- Automated feature engineering and model evaluation pipeline on **100K+ fiber-optic network records**, reducing manual analysis time by ~60% and delivering direct output to the network optimization team.
- Built a **clustering pipeline mapping customer performance profiles** to 5 commercial service tiers using K-Means, enabling data-driven maintenance recommendations for network engineers.
- Developed Random Forest classification models on upload speed, latency, and jitter signals to classify **network quality across thousands of nodes** with Scikit-learn and Pandas.
- Delivered stakeholder presentations translating model outputs into actionable insights, visualized with **Power BI dashboards** for non-technical audiences.

## PROJECTS

### NVIDIA Nemotron Model Reasoning Challenge (In Progress)

Kaggle Competition · Jun 2026 Deadline · \$106K+ Prize Pool

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built diagnostic evaluation pipeline revealing that reasoning over-expansion - not incorrect reasoning - was the primary failure mode, and ran controlled ablation experiments across LR schedules and synthetic augmentation strategies, discovering that reasoning-style consistency outweighs data volume in LLM SFT.

PyTorch · Unsloth · TRL / PEFT · Hugging Face Transformers · vLLM

### AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR CodaBench · 2025

- Ranked **1st in EURECOM class leaderboard** (0.791 AUC) and submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark by fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head and dual learning rates (2e-6 backbone, 5e-4 head).
- Discovered that **1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC)** for out-of-distribution generalization; applied a 3-model ensemble weighted by validation AUC with 10-view TTA across 250K training images and 25+ generator types.

PyTorch · OpenCLIP · ViT · AdamW · FP16 Mixed Precision

## EDUCATION

### Engineering Degree - Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Web Applications, Statistics, Image Security

### CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

## SKILLS

**LLM Fine-Tuning & Generative AI:** LoRA / QLoRA · SFTTrainer · LangChain · RAG · Agentic Pipelines · Prompt Engineering · vLLM

**ML / DL Frameworks & NLP:** PyTorch · Hugging Face Transformers · Scikit-learn · OpenCLIP · TRL / PEFT · Unsloth · TensorFlow

**Data, APIs & MLOps:** Python · Pandas · NumPy · REST APIs · Docker · PostgreSQL · Git · Power BI · FP16/BF16 Precision